

Project Abstract

Boston Image – Image Contrast Enhancement in MATLAB

Overview

This project investigates and compares four widely used contrast-enhancement techniques in digital image processing, applied to a single RGB photograph (*boston night.jpg*). Rather than operating directly on RGB pixel values, the image is converted to the HSV (Hue, Saturation, Value) color space so that brightness/contrast adjustments can be isolated to the Value (V) channel while the Hue and Saturation channels remain untouched, preserving the original color character of the scene.

Problem Statement

Low-light and night-time photographs frequently suffer from poor contrast and compressed dynamic range, obscuring detail in both shadows and highlights. A range of classical enhancement algorithms exist to address this, each with different trade-offs between simplicity, global versus local adaptation, and computational cost. This project provides a direct, reproducible visual and methodological comparison of four such algorithms on the same source image.

Methods Compared

Technique	MATLAB Function	Type	Key Idea
Gamma Correction	$V.^{(1/2)}$	Global, parametric	Non-linear power-law remapping of brightness ($\gamma = 1/2$)
Contrast Stretching	imadjust	Global, automatic	Linearly stretches intensity limits to the full display range
Histogram Equalization	histeq	Global, statistical	Redistributes intensities to flatten the overall histogram
Adaptive Histogram Equalization	adapthisteq	Local, tile-based	Equalizes contrast within local image tiles (CLAHE)

Approach

The workflow: (1) import the source image and convert RGB $\rightarrow$ HSV; (2) split the H, S, and V channels; (3) apply each of the four enhancement techniques to the V channel independently, producing four adjusted V channels; (4) recombine each adjusted V channel with the unmodified H and S channels; (5) convert each result back from HSV $\rightarrow$ RGB and rescale to uint8 for display; and (6) present all four outputs side by side using MATLAB's montage function for direct visual comparison.

Expected Outcome

Four enhanced image variants (imgGamma, imgAdjust, imgEq, imgAdapt) and a single montage figure that allows a qualitative comparison of global versus local, and linear versus non-linear, contrast enhancement strategies on a real-world night-time photograph.

Tools & Environment

MATLAB with the Image Processing Toolbox (for `rgb2hsv`, `hsv2rgb`, `imadjust`, `histeq`, `adapthisteq`, and `montage`).